

Image-Native Language Models: Reading and Writing as Visual Generation

Imagized Language Model Project

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Abstract

Most large language models represent language through symbolic text tokens and treat images of writing as an auxiliary input modality. This paper sketches a different research direction: an image-native language model whose inputs and outputs are visual canvases. The model learns from printed pages, handwritten forms, oracle-bone glyphs, bronze inscriptions, seal forms, and other historical scripts as images. It answers by generating readable images containing modern language and historically grounded glyph panels. We propose a 3090-friendly staged architecture based on visual autoencoding, masked visual language modeling, latent diffusion, and retrieval-augmented glyph evolution.

1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but their central representation is normally a sequence of text tokens. Multimodal models accept images, but many still produce text as the primary output interface. For example, GPT-4 is reported as accepting image and text inputs and producing text outputs; GPT-4o’s API documentation lists text and image input with text output for that model. Llama 3 similarly remains a token-based dense Transformer family, with multimodal capabilities added compositionally.

The Imagized Language Model (ILM) goal is different:

image input $\rightarrow$ visual language state $\rightarrow$ image output.

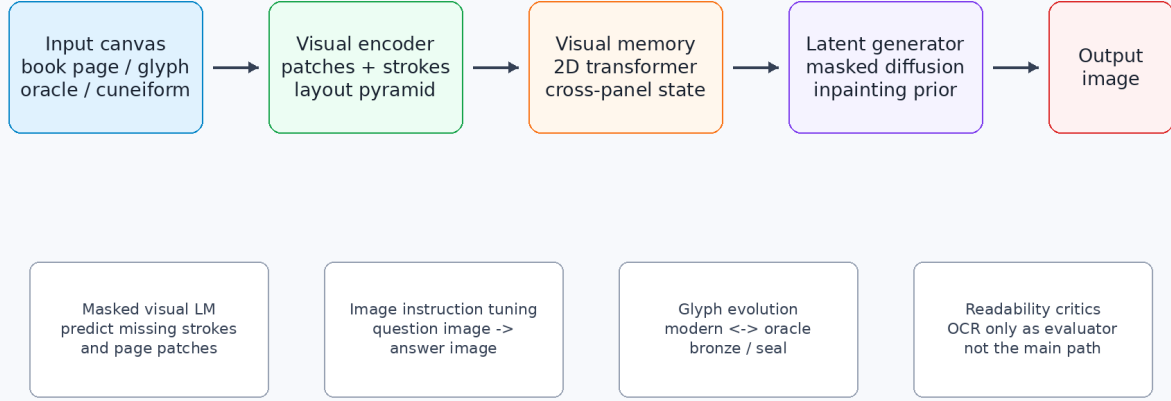
The output should itself be a readable answer image. For a query such as “explain the evolution of character zhong”, the system should produce a page containing modern explanatory writing plus oracle, bronze, seal, and modern forms.

2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. Token-free language models such as CANINE, ByT5, and MEGABYTE show that subword tokenization is not a hard requirement for language modeling. Discrete diffusion language models, including D3PM, Diffusion-LM, SEDD, and LLaDA-style masked diffusion, show that generation need not be left-to-right. Image generative models such as latent diffusion, MaskGIT, and DiT show that high-quality image generation can be performed in compressed latent spaces.

The missing combination is direct: train a model to read and write language as images, while preserving the visual structure of scripts and historical glyphs.

Image-Native Language Model (ILM-V)



Key idea: patches/latents are visual units, not BPE or word tokens.

Figure 1: ILM-V architecture. The model consumes input canvases, encodes them into visual latents, maintains a 2D visual memory, and denoises or fills output image latents. OCR-style readers are only auxiliary evaluators, not the main generation path.

3 Architecture

Figure 1 gives the proposed system. The main components are:

1. **Visual encoder.** A patch/conv hybrid converts glyph tiles, line strips, and page crops into visual latent grids.
2. **Visual memory transformer.** A 2D Transformer maintains cross-panel state with relative position and sparse/window attention.
3. **Latent generator.** A masked diffusion or MaskGIT-like prior fills an output canvas conditioned on input canvas latents.
4. **Image decoder.** A VAE/VQ-VAE decoder converts latents back to readable output images.
5. **Auxiliary readers and critics.** OCR/readability heads evaluate the output, but should not become the primary hidden path.

The internal units are visual patches or visual codebook entries, not BPE, wordpiece, or Unicode language tokens.

4 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

This is enough to train a first glyph autoencoder and a character-evolution generator. Modern book/page data can be added by rendering multilingual corpora into images and by ingesting scanned pages where licenses permit.

5 Training Objectives

The first ILM-V model should combine the following losses:

1. **Masked visual language modeling:** mask image patches and predict missing visual latents.
2. **Denoising reconstruction:** recover pages and glyphs from blur, noise, damage, dropout, and partial occlusion.
3. **Image-to-image instruction tuning:** render prompts and target answers as images and train conditional generation.
4. **Glyph evolution modeling:** map modern forms to historical stages and historical forms back to modern answer sheets.
5. **Contrastive visual semantics:** align different visual renderings of the same content without making OCR the bottleneck.

6 3090-Friendly Curriculum

Figure 2 shows the practical schedule:

1. Build manifests for glyphs, rendered line images, and rendered instruction pairs.
2. Train a 30M–100M parameter glyph/page autoencoder.
3. Train a 100M–250M parameter conditional glyph evolution generator.
4. Train a 200M–600M parameter visual page language model on latent crops.
5. Add instruction tuning with image prompts and image answers.
6. Expand to additional scripts: cuneiform-like signs, hieroglyphic forms, Arabic, Japanese, Korean, and multilingual book pages.

Mixed precision, gradient checkpointing, latent crops, LoRA/adapters, and gradient accumulation make this path realistic on one or two 24GB GPUs.

3090-Friendly Training Curriculum

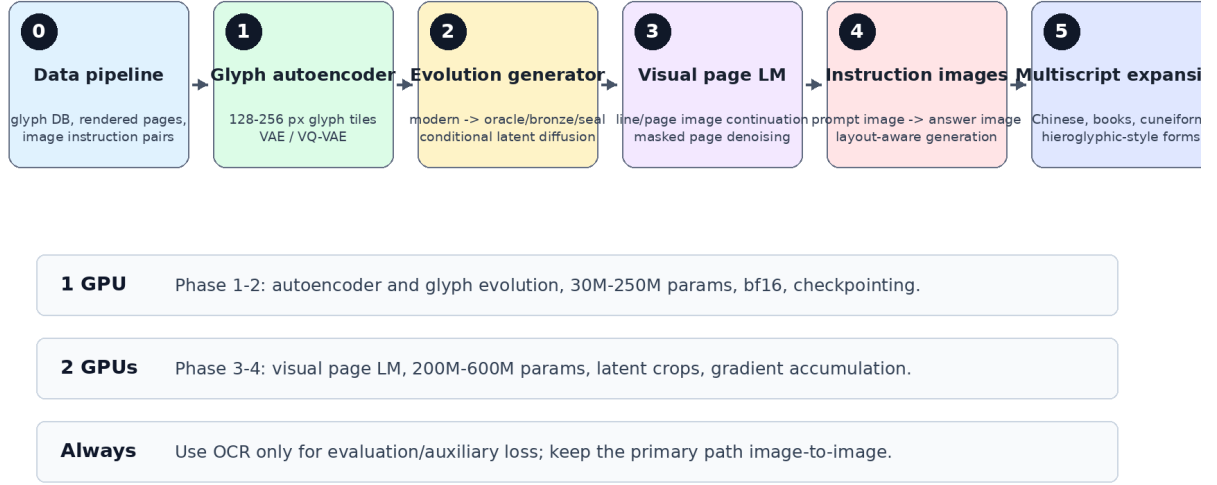


Figure 2: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

7 Target Demonstration

The first demo should answer a rendered visual prompt: “Explain the evolution of the Chinese character zhong.” The target answer image should contain:

- a title and concise modern explanation,
- a historical timeline with real glyph exemplars,
- stage labels and citations,
- modern-language text as rendered image.

This directly tests whether the model can combine language understanding and visual-script generation.

8 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

9 Evaluation

The model should be evaluated as both a language model and a visual generator:

Example Output: Evolution of Chinese Character Zhong (中)

A target answer is itself an image: explanation plus historical forms.

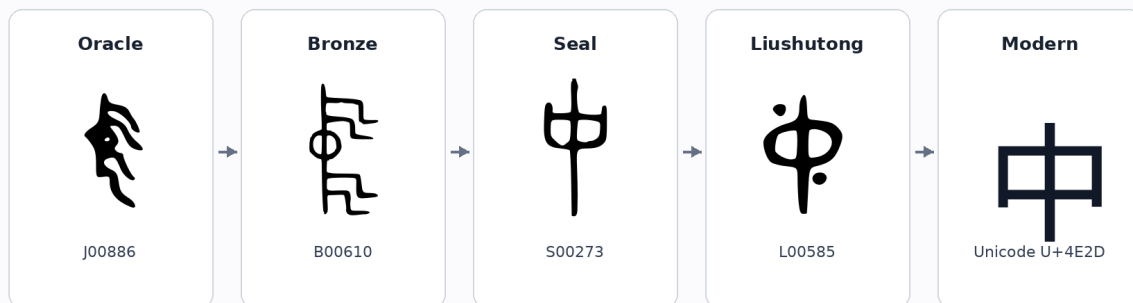


Figure uses local hanziyuan-derived glyph files when available; generation targets should retrieve/cite real exemplars.

Figure 3: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

- OCR readability of generated modern text images, used only as an external metric.
- Human readability and task-success ratings.
- Glyph-stage retrieval accuracy: generated glyphs should retrieve correct characters and stages.
- Historical plausibility: distance to nearest real glyph exemplars in stage-specific embedding space.
- Layout validity: no text/panel overlap, stable reading order, correct answer composition.
- Image-to-image QA accuracy on rendered prompts and rendered answers.

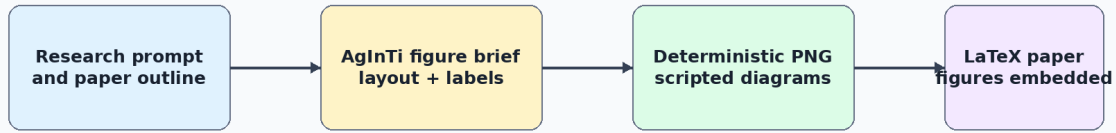
10 Conclusion

ILM-V is a feasible research direction because the pieces exist: OCR-free visual readers, token-free language models, visual latent autoencoders, masked image generators, and diffusion language models. The contribution is to make visible writing itself the modeling substrate. The local Chinese etymology dataset gives a concrete first domain where image output is essential rather than decorative.

References

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AgInTi-Assisted Research Artifact Loop



Why scripted PNGs now?

They are reproducible, editable, and compile cleanly in the paper while leaving room for later AgInTi/image-model refinement

Figure 4: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

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